AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I074*

Team Name: *TEAM MAYRIX*

Domain: *2 - ENERGY, MOBILITY AND SMART INFRASTRUCTURE*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Problem Statement

Aim: Our team aims to solve the problem of unpredictable campus crowd congestion and inefficient mobility management. CampusFlow aims to address this through AI-powered crowd forecasting, shuttle-demand prediction, real-time crowd information, and preventive recommendations to improve campus safety, accessibility, and resource utilisation.

Problem Understanding

- People who experienced the problem are: Students, faculties and staff, campus administrators, transport managers and shuttle operators, campus security and safety teams.

- Cause of the problem: 1. Sudden changes in student movement 2. Fixed shuttle schedules 3. Poor crowd prediction 4. Concentrated demand 5. Fragmented information 6. Reactive management
  Root cause : The lack of integrated, predictive campus crowd and mobility management can make it difficult to anticipate changing demand and take timely action.

- Effects caused to users/stake holder: Wasted time, inefficient resource utilisation, a poorer campus experience, and potential safety concerns.

Existing Gaps & Challenges

- 1. Lack of Predictive Crowd Management: Existing campus systems often rely on current observations or manual monitoring rather than forecasting future crowd levels. This makes it difficult to identify potential congestion early and take preventive action.
- 2. Disconnected Crowd and Mobility Management: Crowd monitoring, shuttle scheduling, and accessibility planning are often handled separately. The lack of an integrated system makes it difficult to predict transportation demand, optimise resources, and provide students with timely information about crowded locations and alternative routes.
- The main challenge that remains unsolved is accurately predicting campus crowd levels and shuttle demand in advance and converting those predictions into timely, practical actions.

Our Proposed Solution

- 1. Core Idea or Product: CampusFlow is an AI-powered campus crowd and mobility management platform that predicts crowd levels, forecasts shuttle demand, and helps campuses manage congestion efficiently.
- 2. How the Solution Works at a High Level: CampusFlow analyses class schedules, campus events, historical crowd data, and shuttle usage to forecast congestion and transportation demand. It provides crowd alerts, suggests alternative routes, predicts busy locations, and recommends better shuttle allocation through a student interface and an administrator dashboard.
- 3. Main Value Delivered to Users: CampusFlow helps students save time, avoid overcrowded areas, and access better travel information. It enables campus administrators to make data-driven decisions, improve shuttle utilisation, manage congestion proactively, and create a safer, more accessible campus experience.

Technical Approach

DATA INPUTS –> DATA PROCESSING –>
AI/ML PREDICTION –> DECISION ENGINE
–> STUDENT AND ADMIN DASHBOARDS

[Explain how the system components work together.]

Tech Stack

Frontend: STREAMLIT, PLOTLY

Backend: PYTHON

Database: PostgreSQL

AI/ML, APIs, or Hardware: NumPy, Pandas; Weather APIs and aggregate crowd-count inputs.

Innovation & Uniqueness

- What is new or different? CampusFlow combines AI-powered crowd prediction, shuttle-demand forecasting, and proactive congestion management in one integrated platform.
- How is the approach suited to the problem? It uses campus schedules, events, and historical crowd data to predict busy locations and recommend timely actions, helping campuses move from reactive management to proactive planning.
- What is the key differentiator? Unlike systems that focus only on crowd monitoring or transportation, CampusFlow connects crowd forecasting, mobility optimisation, accessibility, and actionable recommendations in a single solution.

Feasibility & Scalability

- **Feasibility:** CampusFlow can be developed using Python, Streamlit, Scikit-learn, and PostgreSQL. The initial prototype can use class timetables, campus events, historical crowd counts, and shuttle schedules, without requiring expensive hardware

- **Scalability:** The system can initially be piloted at one canteen and one shuttle stop, then expanded to libraries, academic blocks, and the entire campus. In the future, it can be adapted for other universities and offered as a subscription-based SaaS platform.

- **Risks:** Limited historical data may affect prediction accuracy, while data privacy and integration with existing campus systems may pose challenges. These can be mitigated by starting with manual or aggregate crowd counts, validating predictions during a small pilot, protecting student data, and expanding gradually.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Student Crowd Awareness:
  A student checks CampusFlow before visiting the canteen or library. The system predicts crowd levels and suggests less crowded locations, helping the student save time.

- Use case 2: Shuttle Demand Prediction:
  A campus transport manager views predicted passenger demand at shuttle stops. CampusFlow recommends adjusting shuttle allocation and timings to reduce waiting times and overcrowding.

- Use case 3: Proactive Crowd Management:
  A campus administrator receives alerts about predicted congestion near academic blocks or event venues. The administrator can take preventive action, manage resources, and improve campus safety.

Impacts and Benefits

- Benefits to Target Users: Helps students avoid crowded locations and long queues, while enabling campus administrators to manage crowds, optimise shuttle services, and make informed decisions.
- Social, Economic, and Environmental Impact: Improves campus safety and accessibility, reduces wasted time and inefficient resource usage, and can help reduce unnecessary shuttle trips and fuel consumption.
- Expected Measurable Outcomes: Reduced waiting times, improved shuttle utilisation, more accurate crowd predictions, fewer congestion hotspots, and better student satisfaction. These improvements will be measured during the pilot rather than assumed as proven results.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 11	11.2	Sustainable cities and communities	Improves campus mobility through crowd prediction, better shuttle allocation, and accessible transportation planning.
SDG 9	9.1	Industry, innovation and infrastructure	Uses AI-powered crowd prediction and mobility management to improve campus infrastructure efficiency.

Individual Member Contribution

Member	Contribution or responsibility
Tanishq Rathi	Feasibility Analyst
Manoj Midran M	Presentation Designer
K Anish	Problem Researcher
Arya V	Idea Developer

Yukti ID

IR2026-1051054